

Critical Reflective Piece

Please access my e-portfolio using the following link: <https://gesine-2000.github.io/gesine-linn.github.io/index.html>

You can navigate to this module via the menu bar, or access it directly here: <https://gesine-2000.github.io/gesine-linn.github.io/Research.html>

Introduction

This reflective piece critically examines my learning and development throughout the Research Methods and Professional Practice module, focusing on the development of my statistical analysis skills, my engagement with the research methods process, and the impact of this learning on my personal and professional practice. Using Rolfe et al.'s (2001) WHAT–SO WHAT–NOW WHAT framework, this reflection synthesises evidence from my e-portfolio while critically analysing how my prior experience, emotional responses, and assumptions shaped my learning. Rather than describing module activities, this reflection focuses on moments of challenge and reassessment that significantly altered how I approach research and evidence-based decision-making (Moon, 2004).

WHAT? – Learning Context and Critical Incidents

Entering the module, I already had academic and practical exposure to data-driven work. My Bachelor's thesis focused on project controlling, where I analysed cost, time, and performance indicators, and I had previously developed a Power BI report to visualise project data for managerial decision-making. As a result, I initially assumed that my existing analytical background would transfer easily to academic research methods. However, early reflective activities on ethics and collaborative discussions challenged this assumption by revealing that methodological rigour involves more than technical competence (Biggs and Tang, 2011).

A key learning moment occurred during the statistics workshops and mandatory worksheets in Units 8 and 9. While I was comfortable generating descriptive outputs and visualisations, I struggled when required to justify methodological choices and

interpret uncertainty. This created frustration and self-doubt, as I realised that my prior experience prioritised useful outputs over methodological justification. Similarly, developing my literature review and research proposal exposed a tendency to over-scope research questions, reflecting a project-oriented mindset rather than a research-driven one. Peer review activities and group discussions further highlighted differences, prompting reflection on my own contributions (Brookfield, 2017).

SO WHAT? – Critical Analysis and Interpretation

Development of Statistical Analysis Skills

The most significant shift in my statistical learning was moving from instrumental use of data to critical interpretation. My prior work had conditioned me to value clarity, dashboards, and actionable metrics. While valuable in professional contexts, this mindset initially limited my appreciation of statistical assumptions, uncertainty, and validity. Through the module's inferential statistics exercises, I became more aware of how easily numerical outputs can be misleading if assumptions are violated or context is ignored (Field, 2018).

Emotionally, this was challenging, as it undermined my confidence in skills I previously considered strengths. I recognised a tendency to trust tools and visualisations without sufficiently questioning data quality or sampling limitations. This aligns with Gigerenzer et al.'s (2007) critique that statistical tools can create an illusion of certainty. Reconstructing datasets independently further reinforced that data preparation and validation are integral analytical skills, not preliminary technical steps. As a result, I now approach statistical analysis with greater caution, prioritising interpretation over presentation.

Reflection on the Research Methods Process

The module fundamentally changed how I understand the research process. Initially, I approached research with a project-management mindset, assuming that defining objectives early would ensure efficiency and clarity. In practice, iterative engagement with the literature revealed misalignments between my research aims, methodological choices, and available data. This forced me to confront the limitations of linear planning in academic research (Creswell and Creswell, 2018).

Looking back, my initial research proposal tried to cover too much and was not realistic in terms of what could be achieved. This reflects my professional background in managing projects, where broad objectives are often encouraged, but in an academic context this led to a lack of methodological focus. Feedback and peer review helped me recognise the need to narrow my research questions and better align them with appropriate methods and practical limitations. Ethical discussions further shifted my understanding by showing that ethics is not just a formal requirement, but something that influences research decisions throughout the entire process (Saunders et al., 2019).

Personal and Professional Impact

From a personal and professional perspective, the module exposed both strengths and limitations in my skill set. While my background in project management and data visualisation provided a strong foundation, the professional skills matrix and SWOT analysis revealed gaps in advanced statistical reasoning and academic methodological confidence. Acknowledging these gaps was uncomfortable, as it conflicted with my self-perception as a competent data-oriented professional.

However, this awareness has positively reshaped my professional practice. I now question metrics, dashboards, and analytical outputs more critically, recognising the assumptions embedded within them. This shift is particularly relevant for professional environments where data-driven decisions can have strategic implications.

Developing comfort with uncertainty has improved my ability to communicate limitations transparently, an essential skill in complex organisational contexts (Schön, 1983).

NOW WHAT? – Changed Thinking and Future Actions

As a result of this module, my approach to research and analysis has fundamentally changed. I now begin analytical tasks by critically evaluating assumptions, data quality, and methodological fit rather than focusing on outputs alone. For my MSc dissertation, I will adopt a more iterative research design process, using early literature engagement to refine research questions and ensure coherence between theory and method.

Professionally, my long-term goal is to develop into an international IT project manager, where decisions are frequently made across cultural, organisational, and regulatory contexts. The module has highlighted that effective project leadership requires not only technical and managerial skills but also the ability to critically evaluate evidence and uncertainty. To support this goal, I plan to further develop my skills through targeted training and applied practice, as outlined in my action plan. Reflective practice will remain central to my development, enabling continuous learning in dynamic professional environments (Kolb, 1984).

Conclusion

Overall, this module transformed my understanding of research from a task-focused extension of project work into a critically reflective practice. By challenging my assumptions, exposing limitations, and integrating methodological reasoning with professional experience, the module has strengthened both my academic capability and my readiness for evidence-based decision-making in an international IT project management context.

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References

- Biggs, J. and Tang, C. (2011) *Teaching for Quality Learning at University*. 4th edn. Maidenhead: Open University Press.
- Brookfield, S.D. (2017) *Becoming a Critically Reflective Teacher*. 2nd edn. San Francisco: Jossey-Bass.
- Creswell, J.W. and Creswell, J.D. (2018) *Research Design: Qualitative, Quantitative, and Mixed Methods Approaches*. 5th edn. London: Sage.
- Field, A. (2018) *Discovering Statistics Using IBM SPSS Statistics*. 5th edn. London: Sage.
- Gigerenzer, G., Gaissmaier, W., Kurz-Milcke, E., Schwartz, L.M. and Woloshin, S. (2007) 'Helping doctors and patients make sense of health statistics', *Psychological Science in the Public Interest*, 8(2), pp. 53–96. Available at: <https://doi.org/10.1111/j.1539-6053.2008.00033.x>
- Kolb, D.A. (1984) *Experiential Learning: Experience as the Source of Learning and Development*. Englewood Cliffs, NJ: Prentice Hall.
- Moon, J.A. (2004) *A Handbook of Reflective and Experiential Learning*. London: RoutledgeFalmer.

Rolfe, G., Freshwater, D. and Jasper, M. (2001) *Critical Reflection in Nursing and the Helping Professions*. Basingstoke: Palgrave Macmillan.

Saunders, M., Lewis, P. and Thornhill, A. (2019) *Research Methods for Business Students*. 8th edn. Harlow: Pearson.

Schön, D.A. (1983) *The Reflective Practitioner*. New York: Basic Books.